

May 6 lecture : Introduction to Ricci flow

Riemannian manifolds

M^n smooth manifold

Riemannian metric — infinitesimal notion
of distance & angle.

(g)

Riemannian geometry.

$g \mapsto \nabla$ Levi-Civita connection.

$\mapsto (Rm)$ Riemann curvature 4-tensor.

measures noncommutativity of ∇ .
trace (averaging).

$\xrightarrow{\quad} Ric$ Ricci tensor

Hamilton's Ricci flow.

(M^n, g_0) initial metric.

Search for canonical or good geometric structures

Heat-type equation for metrics:

Fix M^n , Consider 1-parameter of metrics
 $g(t)$ with $g(0) = g_0$, $t \in [0, \omega)$.

Met g_0 $g(t)$

Hamilton's eqn:

$$\partial_t g(t) = -2 Ric_{g(t)}$$

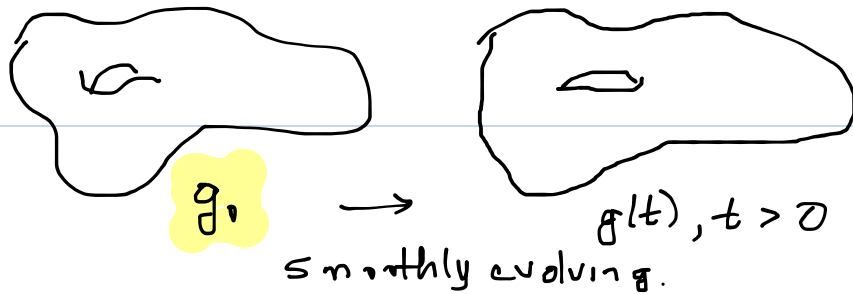
This is a geometric PDE.

Philosophy: The geometry sharp.

Short-time existence thm (Hamilton, DeTurck).

M^n compact, $g_0 \in C^\infty$

Then there exists a unique ^{smooth} solution to the RF $g(t)$ with $g(0) = g_0$ defined on some time interval $[0, \epsilon)$, $\epsilon > 0$.



$g(t)$, as $t \uparrow$, is getting better.

Examples:

$(M^n, g_0) = n$: sphere of radius $r_0 > 0$.



Under RF, stays round, shrinks to a point.

in some finite time $T_0 = \frac{r_0^2}{2(n-1)}$.

time $\approx$ distance square.

$$\text{RF eqn} \Rightarrow \frac{dr}{dt} = -\frac{n-1}{r}$$

$$r(t) = \sqrt{r_0^2 - 2(n-1)t}$$

$$\text{Einstein metric } Ric_0 \equiv c_0 g_0 \Rightarrow$$

$$g(t) = c(t) g_0, \quad c(t) > 0.$$

$c_0 > 0 \Rightarrow$ Einstein metric shrinks homothetically to a point in finite time.

$c_0 = 0 \Rightarrow$ static Ricci flat.

$c_0 < 0 \Rightarrow$ Einstein, expands homothetically.

The geometries scale but don't really change. After scaling, they are fixed points of the Ricci flow.

$$n=2: \quad \text{Ric} = \frac{1}{2} R g, \quad R = \text{scalar curvature}$$

RF: $\partial_t g = -R g$. same as the Yamabe flow.

solution $g(t) = e^{2u(t)} g_0$ $u(t)$ function.

2D: RF is a conformal flow.

Standard:

$$R_{g(t)} = e^{-2u(t)} (R_{g_0} - 2 \Delta_{g_0} u(t)).$$

$$\text{RF} \Leftrightarrow \frac{\partial u}{\partial t}(t) = e^{-2u(t)} \Delta_{g_0} u(t) - e^{-2u(t)} \frac{R_{g_0}}{2}.$$

one can study this heat-type eq for $u(t)$

Hamilton's approach was geometric:

$$\circ \text{Normalized RF: } \partial_t g = (r - R) g$$

$$\text{where } r = \frac{\int R dV}{\int dV} = \text{constant} \Rightarrow \int dV = \text{const.}$$

Hamilton gave RF proof of uniformization thm (geom. version).

$$V(M^2, g_0) \Rightarrow \exists \text{ normalized RF } g(t), t \in [0, \infty)$$

& $g(t) \rightarrow g_\infty$ metric on M^2 where $R_{g_\infty} \equiv \text{const.}$

• Estimate for R :

Solved $\Delta f = R - r$
 $S = 0$.

$$\Rightarrow \frac{\partial f}{\partial t} = \Delta_g f + r f$$

RF is more linear than you think.

roughly estimate

$$H = R + |\nabla f|^2 - r$$

$$\partial_t R = \Delta R + R^2 - rR \quad \text{semilinear.}$$

causes problems.
 bad term.



$$\star \frac{\partial}{\partial t} |\nabla f|^2 = \Delta |\nabla f|^2 - 2 |\nabla^2 f|^2 + r |\nabla f|^2$$

$\times$ good $\nabla \nabla f$ linear.
 $|\nabla x|^2$

& better than |bad|.

$$\partial_t H = \Delta H - 2|M|^2 + rH \quad \leftarrow \text{sharp.}$$

$\nwarrow$ linear

where

$$M = \nabla^2 f - \frac{1}{2} \Delta f \cdot g \rightarrow \text{trace-free part of the Hessian of } f$$

$$= \nabla^2 f - \frac{1}{2} (R - r) g$$

Comment: $M = 0 \Leftrightarrow$ "shrinking gradient Ricci soliton".

has geometric significance.

Key ingredients.

- Harnack estimate.

- Li-Yau inequality — 1986.

- parabolic version Yau / Cheng-Yau.

- gradient estimates for Laplace equation.

assume $Ric \geq 0$ to get cleanest.

harmonious with Bishop-Gromov volume comparison

- Entropy estimate:

$$\int_M R \log R / A dV. \quad \text{assume } R > 0.$$

monotone under 2D RF.

- Soliton estimate.

$$M = \underbrace{\nabla^2 f - \frac{1}{2} \Delta f g}_{\frac{1}{2} L_{\nabla f} g} = \text{modified scalar curvature.}$$

$$\Delta f = R - r.$$

$$\left[\frac{\partial}{\partial t} |M|^2 = \Delta |M|^2 - 2 |\nabla M|^2 - 2 R |M|^2 \right]$$

Consider $M^2 = S^2$ ^{assume} $R > 0$. a good equation.

$g(t)$ after diff, $\rightarrow g_\infty \leftarrow$ shrinking gradient Ricci Soliton.

$\Rightarrow$ NRF converges to const. curvature

constant curvature

w/o unif:

Chen, Lu, Tian.

used uniformization theorem